# Xujiang Tang

Theoretical Mathematics Researcher; Machine Learning Theory Researcher
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Visiting Student, Great Bay University
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## RESEARCH INTERESTS

- Algebraic topology and equivariant homotopy theory: fixed point methods, geometric fixed point tomography, Bokstedt periodicity, and stable homotopy-theoretic structures.
- Finite group theory: Quillen's p-subgroup conjecture, subgroup complexes, centralizer structures, and FRUL-WOS sphere methods.
- Formal proof: Lean, Mathlib, and formalization of advanced mathematics in algebra and topology.
- Machine learning theory: generalization, learnability, out-of-distribution prediction, and Koopman/operator-theoretic methods for long-sequence data.

## COLLABORATORS AND MENTORS

- William Redman, Johns Hopkins University - https://engineering.jhu.edu/faculty/william-redman/
- Ziyue Qiao, Great Bay University - https://qiaoziyue.com/

## EDUCATION

- Yangtze University College of Arts and Sciences, Jingzhou, China. B.Sc. in Mathematics and Applied Mathematics, expected June 2027.
- Selected coursework: mathematical analysis, linear algebra, probability and statistics, differential equations, computational modeling, optimization theory, numerical computation, data structures, Python, R, and MATLAB.
- Selected marks: Optimization Theory 97/100; Numerical Computation Methods 96/100.

## VISITING POSITION AND TRAINING

- Visiting Student, Great Bay University, 2025-present. Research in theoretical mathematics and machine learning theory.
- Zhejiang University Lean Formalization Summer School, 2026. Formal proof, Lean, Mathlib, and computer-assisted theorem proving.
- Neural Networks and Python Implementation, short winter training with Prof. Mark Vogelsberger. Grade: A.

## RESEARCH EXPERIENCE

- Finite Group Theory and Quillen's p-Subgroup Conjecture, 2025-present. Work on centralizer energy, FRUL-WOS spheres, and subgroup-complex methods related to Quillen's p-subgroup conjecture.
- Algebraic Topology and Equivariant Homotopy Theory, 2025-present. Work on geometric fixed point tomography and spoke Bokstedt periodicity.
- Epidemic Prevention and Population Mobility Modeling, Jingzhou, 2023-2025. Worked with Prof. Qingpeng Ran, Department Chair, on mathematical modeling and statistical surveys related to epidemic prevention and population mobility in Jingzhou.
- Spatial Omics Data Processing, 2025-2026, first half. Worked with Dayu Hu, Northeastern University, on data processing for spatial omics.
- Epilepsy and Long-Sequence Data Research, 2025-2026, second half. Studied epilepsy-related data and long-sequence modeling in Prof. Huaguang Gu's group, focusing on Koopman operator methods.
- Formal Proof and Lean, 2026-present. Study of Lean formalization, proof engineering, and the long-term formalization of advanced mathematical arguments.
- Theoretical Machine Learning and Optimization, 2024-present. Work on learnability, generalization guarantees, out-of-distribution prediction, and optimization-theoretic aspects of learning algorithms.
- Stochastic Processes, Dynamical Systems, and Quantitative Finance, 2023-present. Research on stochastic-process models, rough volatility, MCMC-based simulation, Koopman/operator-theoretic dynamics, and robust financial decision problems.

## PUBLICATIONS AND MANUSCRIPTS

Submitted manuscripts:

- X. Tang. Centralizer Energy, FRUL-WOS Spheres, and Quillen's p-Subgroup Conjecture. Submitted to Annals of Mathematics.
- X. Tang. Geometric Fixed Point Tomography and Spoke Bokstedt Periodicity. Submitted to Algebraic & Geometric Topology.
- X. Tang and C. Fan. Out-of-Distribution Generalization Error Bounds for Successful Prediction of Deep Autoregressive Algorithms. Submitted to NeurIPS 2026.
- X. Tang. Quartic Difficulty: Assessing the Learnability of Unsupervised Learning Algorithms.

Submitted to COLT.

Published and accepted papers:

- Q. Li and X. Tang. HKGEduRec: A Knowledge Graph-Enhanced Dynamic Hybrid Framework for Educational Recommendation with Cold-Start Mitigation. IEEE ICMEIM 2025. EI-indexed.
- X. Tang and Q. Li. Logical Gene Encoding: A Bio-Inspired Approach for Energy-Efficient Automated Reasoning. IEIT 2025. EI-indexed. DOI: 10.2991/978-94-6463-803-5_81.
- Q. Li and X. Tang. Robust Optimal Reinsurance and Investment with Inflation Risk: A Game-Theoretic Approach and Explicit Solutions. AIMS Mathematics. Accepted, 2025.

## SOFTWARE COPYRIGHTS

- Software copyright in financial systems.
- Software copyright in unmanned swarm systems.

## HONORS, COMPETITIONS, AND ACTIVITIES

- Kaggle Silver Medal, Jigsaw Toxic Comment Classification Challenge.
- Regional Gold Medal, WorldQuant International Quant Championship.
- Third Prize, UCAS Graduate AI Forum.
- Mathematical Modeling Competition, Hua Zhong Cup.
- National College Students Statistical Modeling Competition.
- Multiple national second prizes in mathematical modeling competitions.
- Reviewer, ICLR 2026 Workshop.

## PROFESSIONAL EXPERIENCE

- Peking University - Open-Source LLM Group, Jun.-Aug. 2024. Contributed to LLM deployment and retrieval-augmented generation systems.
- Deloitte, Jul.-Sep. 2024. Professional experience in consulting workflows, teamwork, and communication.
- Guolian Securities, Jul.-Sep. 2024. Applied statistical methods to securities research and quantitative workflows.

## SKILLS

- Mathematics: algebraic topology, finite group theory, homotopy theory, representation theory, optimization, numerical analysis, stochastic processes, differential equations, causal inference, statistical modeling.
- Formal proof and programming: Lean, Mathlib, Python, PyTorch, NumPy, Pandas, MATLAB, R, C++, LaTeX, Git/GitHub, Linux.
- Languages: Mandarin Chinese, English.